

My Research Interest

...it's quite involved, so I'll get to the title later, promise

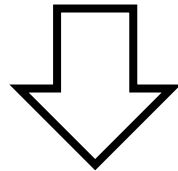
Overview

1. How do Websites work?
 - a. HTML = Layout
 - b. CSS = Style
 - c. JavaScript = All the cool stuff
 - d. Modern Website Development
2. What is a Fluent DSL?
3. Making Fluent DSLs Secure

How do Websites Work?

HTML = Layout

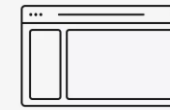
```
<div>Hello World!  
  <br>  
  <button>Click me!</button>  
</div>
```



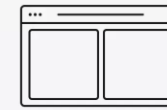
Hello World!

Click me!

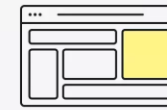
Types of Web Design Layout



Two-column
Layout



Split Screen
Layout



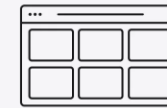
Asymmetrical
Layout



The F-shape
Layout



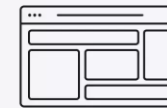
The Z-shape
(zig-zag) Layout



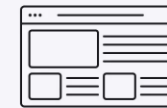
Card or Block
Layout



The Featured
Image or Video
Layout



Masonry
Layout



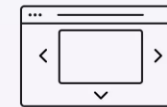
Magazine
Layout



Layout with
Fixed
Navigation

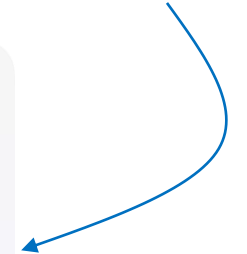


Layout with
Hidden
Navigation



Interactive
Layout

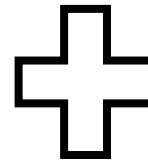
specified by
HTML



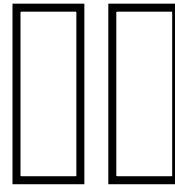
How do Websites Work?

CSS = Style

```
<div>Hello World!  
  <br>  
  <button class='clickable'>Click me!</button>  
</div>
```



```
.clickable {  
  background-color: ● rgb(5, 5, 177);  
  color: ● white;  
  border-radius: 2px;  
  border: none;  
  padding: 5px;  
}  
  
.clickable:hover {  
  background-color: ● rgb(5, 5, 100);  
}
```



Hello World!

Click me!

Hello World!

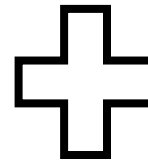
Click me!



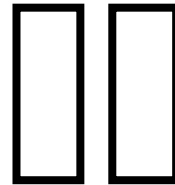
How do Websites Work?

JavaScript = All the cool stuff

```
<div>Hello World!  
  <br>  
  <button class='clickable' onclick='sayHello()'>Click me!</button>  
</div>
```



```
function sayHello() {  
  console.log("Hello!");  
}
```



```
"Hello!"
```

```
>_
```

Modern Website Development

- Every Language covers its own **problem domain**:
 - HTML = Layout
 - CSS = Style
 - JavaScript = Behaviour

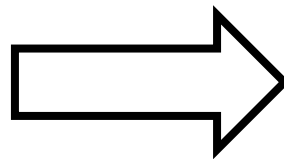
→ Very Complex: Developers need to be comfortable with 3 language
(at least!)

Solution: Fluent DSLs

What are Fluent DSLs?

- DSL = **Domain Specific Language**
- Fluent DSL = A DSL using familiar syntax

```
HTMLElement.builder()
  .tag("div")
  .text("Hello")
  .addChild(HTMLElement.builder()
    .tag("br")
    .text("World!")
    .build())
  .build()
```

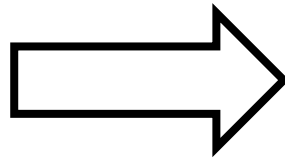


```
<div>Hello  
  <br>  
  |   World!  
  </br>  
</div>
```

My Research Goal

- Prevent misuse of Fluent DSLs:

```
HTMLElement.builder()  
  .tag("div")  
  .tag("input")  
  .text("Hello")  
  .build()
```



double tagging
is not allowed

My Research Goal

- Research Title: Static Correctness for Fluent DSLs
- Current Progress:
 - Identified host language: Java
 - Identified DSL to embed: SQL

Conclusion

- Modern Software uses a mix of programming languages
- This is a strain on Developers
- Fluent DSLs alleviate this
- But come with their own problems!
 - No safety guarantees
 - Feature incompleteness

Use Case 1: Smart Error Messages

```
Query query = SQL.select("*")  
                .from("Students")  
                .from("Teachers")  
                .end();
```

```
Query query = SQL.select("*")  
                .from("Students")  
                .from("Teachers")  
                .end();
```

Expected 'where' or 'end' but got 'from'

Use Case 2: Analyzing String Literals

```
Query query = SQL.select("1 + 5")  
                .from("Students")  
                .end();
```

```
Query query = SQL.select("1 + 5")  
                .from("Students")  
                .end();
```

Expected '*' or ID but got '1'